

Euclid’s Elements — Book XI

Proposition 20

Formal Reconstruction — Technical Documentation

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If a solid angle is contained by three plane angles, any two of them are together greater than the remaining one.

1 Purpose and scope

Proposition XI.20 is the first proposition in the present reconstruction where Euclid’s verbal comparison of a *sum of two plane angles* with a third angle must be represented without introducing a numerical angle measure. The formal development therefore separates three layers:

1. a neutral language for a proper trihedral configuration and for the relation “two angles together are greater than a third”;
2. a planar metric core, based on congruence, SAS, Euclid I.20, and Euclid I.25;
3. one small spatial incidence step which prevents a certain auxiliary triangle from degenerating.

Two production proofs are provided. The direct Hilbert proof uses the three-dimensional incidence interface. The Hilbert–Wyler proof replaces the single spatial step by the dimension-free point–line–plane incidence fragment of `HilbertWylerAxioms`. The two proofs deliberately share the neutral metric core but not the final theorem itself.

2 Classical configuration

Let O be the vertex of a solid angle bounded by the three rays OA , OB , and OC . The three face angles are

$$\angle AOB, \quad \angle BOC, \quad \angle COA.$$

Euclid XI.20 asserts the three cyclic strict inequalities

$$\begin{aligned}\angle AOB + \angle BOC &> \angle AOC, \\ \angle BOC + \angle COA &> \angle BOA, \\ \angle COA + \angle AOB &> \angle COB.\end{aligned}$$

The formal theorem assumes that each face angle is proper and that O, A, B, C do not lie in one plane. This is exactly the content of the neutral predicate `HilbertTrihedralConfiguration`.

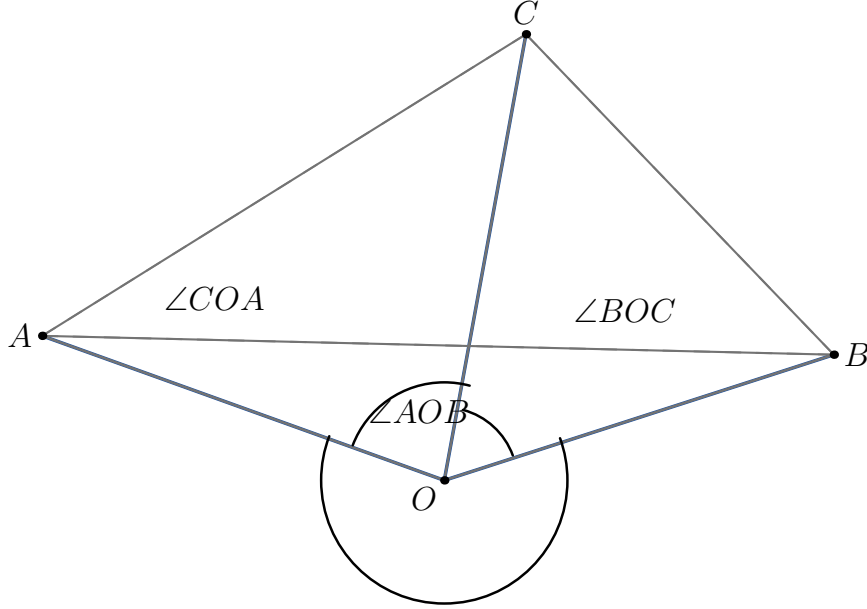


Figure 1: A proper trihedral configuration with vertex O and face angles $\angle AOB$, $\angle BOC$, and $\angle COA$. The drawing is schematic: the essential spatial condition is that O, A, B, C are not coplanar.

3 Neutral synthetic vocabulary

The file `CGJteamLab/HilbertTrihedralAngle.lean` introduces the vocabulary needed by both formal routes.

3.1 Trihedral configuration

The definition

```
def HilbertTrihedralConfiguration
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (O A B C : Geo.Point) : Prop :=
  Not (PrimCollinear Geo A O B) /\
  Not (PrimCollinear Geo B O C) /\
  Not (PrimCollinear Geo C O A) /\
  Not (exists pi : S.Plane,
    S.OnPlane O pi /\
    S.OnPlane A pi /\
```

```
S.OnPlane B pi /\
S.OnPlane C pi)
```

records exactly the three proper face angles and the noncoplanarity of the four points. No orientation of space and no numerical angle are involved.

3.2 What “two angles together are greater” means

The relation `HilbertTwoAnglesGreaterThanAngle` is the central design choice of XI.20. It represents

$$\alpha + \beta > \gamma$$

without defining an addition operation on angles. The target angle γ is handled by one of three mutually sufficient geometric situations:

1. $\gamma < \alpha$ already;
2. $\gamma \cong \alpha$, while β is a proper angle;
3. an interior ray decomposes γ into two components, the first congruent to α , and the remaining component strictly smaller than β .

Thus the third case captures Euclid’s actual construction: copy the first summand into the target angle and compare the remaining part with the second summand. The public predicate `HilbertTrihedralAngleInequalities` is simply the conjunction of the three cyclic instances of this relation.

This representation is important mathematically. XI.20 is proved entirely in synthetic angle order; no real-valued angle measure, no angle addition, and no trigonometry are introduced merely to state the proposition.

4 The classical constructive core

It is enough to prove one cyclic inequality, for example

$$\angle ZOX + \angle ZOY > \angle XOY.$$

Compare the first summand $\angle ZOX$ with the target $\angle XOY$. There are three cases.

If $\angle XOY < \angle ZOX$, the first summand is already larger than the target. If the two angles are congruent, the proper second summand makes the sum strictly larger. The only nontrivial case is

$$\angle ZOX < \angle XOY.$$

By the synthetic definition of angle order there is an interior ray OW of $\angle XOY$ such that

$$\angle ZOX \cong \angle XOW.$$

Let the ray OW meet XY at H . On the third ray OZ choose D so that

$$OD \cong OH.$$

Transporting the copied angle from the rays OW and OZ to the same rays represented by OH and OD gives

$$\angle DOX \cong \angle XOH.$$

Together with $OD \cong OH$ and the common segment OX , SAS yields

$$DX \cong HX.$$

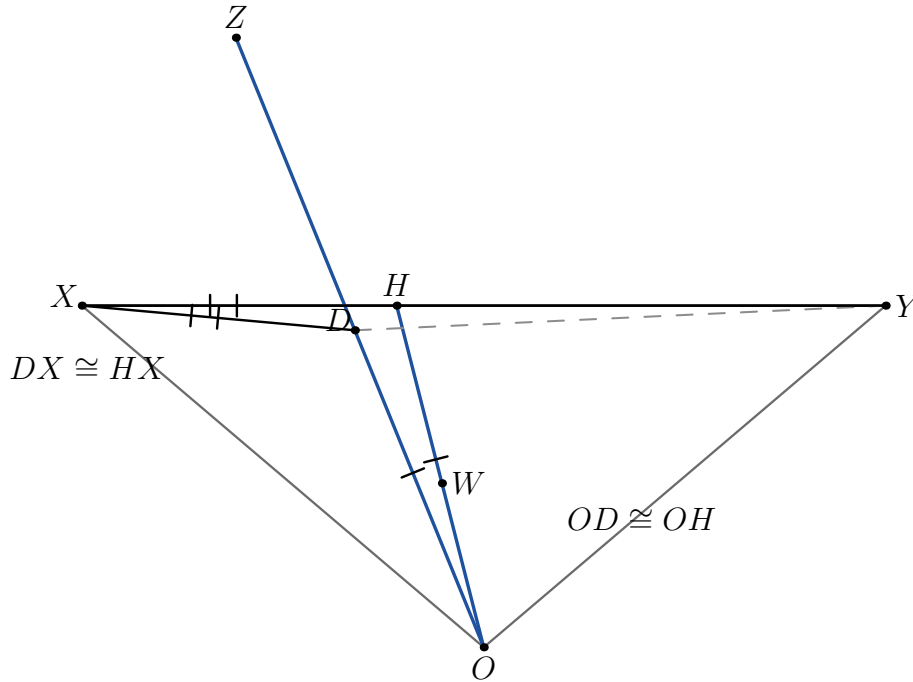


Figure 2: The auxiliary construction used in the nontrivial comparison case. H is the intersection of the interior ray with XY ; D is laid off on OZ with $OD \cong OH$. SAS gives $DX \cong HX$.

Now consider triangle DXY . Since H lies between X and Y and $DX \cong HX$, Euclid I.20 together with segment cancellation gives

$$HY < DY.$$

Applying Euclid I.25 to triangles OHY and ODY , with $OH \cong OD$ and OY common, gives

$$\angle HOY < \angle DOY.$$

Finally, same-ray transport returns from H to W and from D to Z :

$$\angle WOY < \angle ZOY.$$

This is precisely the strict estimate required for the remainder of the target angle.

5 Shared planar metric core

The file `CGJteamLab/HilbertTrihedralAngleMetric.lean` contains the parts of the proof that are independent of the ambient spatial incidence system. Its principal lemmas are:

- `hilbert_XI20_two_angles_greater_of_first_less`: extracts the interior-ray decomposition from the hypothesis that the first summand is smaller than the target;

- `hilbert_XI20_sas_base_congruent`: the SAS step yielding the congruence of the auxiliary base segments;
- `hilbert_XI20_remainder_side_less`: Euclid I.20 plus cancellation, producing the strict side inequality for the remainder;
- `hilbert_XI20_remainder_angle_less`: Euclid I.25, converting that side inequality into the strict angle inequality;
- `hilbertTwoAnglesGreaterThanAngle_swap_first`: symmetry needed to present the final cyclic inequalities in the conventional order of rays.

This factorization is not merely a code refactor. It identifies the genuinely planar metric content of XI.20 and makes the spatial dependency visible as a single separate obligation.

6 The genuinely spatial obligation

The auxiliary application of Euclid I.20 requires a proper triangle DXY . Thus one must prove

$$D, X, Y \text{ are not collinear.}$$

This is the only point in the core construction where the ambient point–line–plane structure is used.

Assume, for contradiction, that D, X, Y are collinear. Since O, X, Y are noncollinear, they determine a plane π . The line XY lies in π ; therefore $D \in \pi$. Since $O, D \in \pi$ and $O \neq D$, the whole line OD lies in π . But O, D, Z are collinear because D lies on the ray OZ . Hence $Z \in \pi$, contradicting the noncoplanarity of O, X, Y, Z .

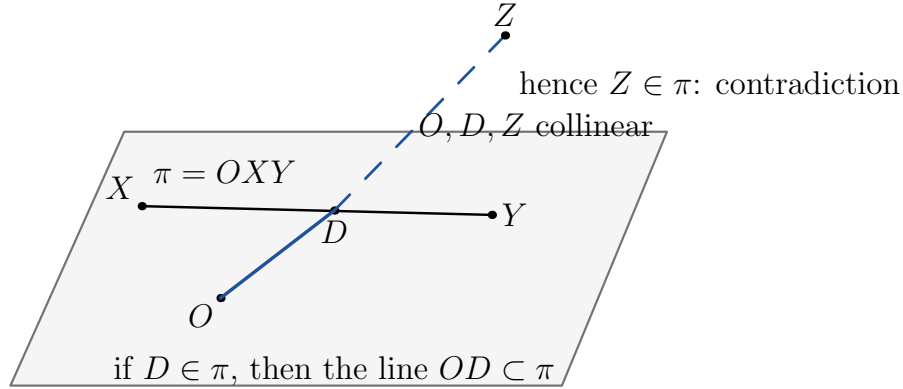


Figure 3: The sole spatial contradiction in the XI.20 construction. If D were on XY , then D would lie in the plane $\pi = OXY$. The line $OD = OZ$ would then lie in π , forcing $Z \in \pi$ and contradicting the trihedral noncoplanarity.

7 Direct Hilbert 3D reconstruction

The direct production theorem is

`theorem euclid_proposition_11_20`

```

[HilbertIncidence Geo]
[S : HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertCongruence Geo]
(O A B C : Geo.Point)
(hTri : HilbertTrihedralConfiguration Geo O A B C) :
HilbertTrihedralAngleInequalities Geo O A B C

```

from CGJteamLab/Proposition11_20.lean.

Its local nontrivial theorem is `hilbert_XI20_trihedral_remainder_less`. In the spatial block the proof uses

```

HilbertSpaceIncidence.plane_through
HilbertSpaceIncidence.line_in_plane

```

to carry out the contradiction described above. Once the auxiliary triangle DXY is known to be proper, the proof returns entirely to the shared metric core.

The theorem `hilbert_XI20_one_cyclic_inequality` then removes the comparison hypothesis by trichotomy between the first summand and the target. The final theorem obtains the remaining two inequalities by permuting the three rays of the trihedral configuration.

8 Hilbert–Wyler reconstruction

The independent production theorem is

```

theorem euclid_proposition_11_20_wyler
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertCongruence Geo]
  [A : HilbertWylerAxioms Geo]
  (O A B C : Geo.Point)
  (hTri : HilbertTrihedralConfiguration Geo O A B C) :
  HilbertTrihedralAngleInequalities Geo O A B C

```

from CGJteamLab/Wyler/Proposition11_20.lean.

The Wyler proof does not call the direct Hilbert theorem. It has its own remainder lemma `hilbert_XI20_trihedral_remainder_less_wyler` and its own cyclic theorem `hilbert_XI20_one_cyclic_inequality_wyler`. Both reuse only the neutral metric lemmas.

The decisive observation is that the spatial contradiction does not require Wyler I.7. It uses only the two fields

```

A.plane_through
A.line_in_plane

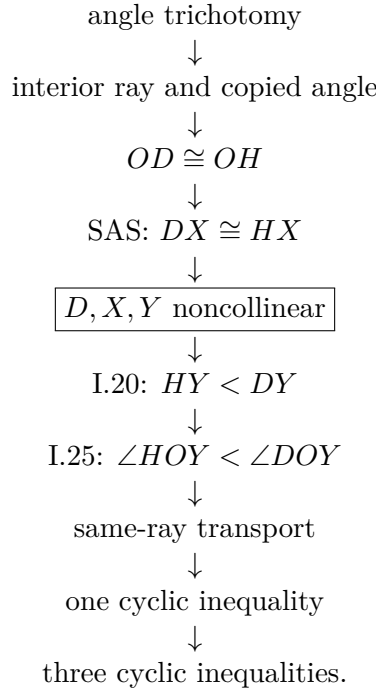
```

of the common Hilbert–Wyler incidence core. No use is made of Smith I.5, span exchange, or an ambient dimension bound.

A small local compatibility instance exposes the `HilbertPlaneIncidence` parent already contained in `HilbertCongruence`; this is only a typeclass-resolution adapter and adds no geometric assumption.

9 Hilbert versus Hilbert–Wyler proof DAG

The two production DAGs coincide after the spatial nondegeneracy step:



The boxed node is implemented differently:

Hilbert 3D	Hilbert–Wyler
<code>HilbertSpaceIncidence.plane_through</code>	<code>A.plane_through</code>
<code>HilbertSpaceIncidence.line_in_plane</code>	<code>A.line_in_plane</code>

Everything below that node is common metric geometry.

10 E^n transfer audit

XI.20 gives an unusually clean answer to the dimensional-transfer question.

1. Purely planar step. The copied-angle construction, SAS, Euclid I.20, Euclid I.25, angle order, and same-ray transport are planar metric geometry. They are already isolated in `HilbertTrihedralAngleMetric.lean`.

2. Genuine spatial incidence step. Only the proof that D, X, Y are noncollinear leaves the planar metric layer. Its logical form is: construct the plane OXY , place D in it from collinearity with X, Y , then place Z in it from collinearity with O, D .

3. Wyler replacement. The replacement needs only “three noncollinear points lie in a plane” and “a plane containing two distinct points of a line contains that line.” These are already fields of `HilbertWylerAxioms`.

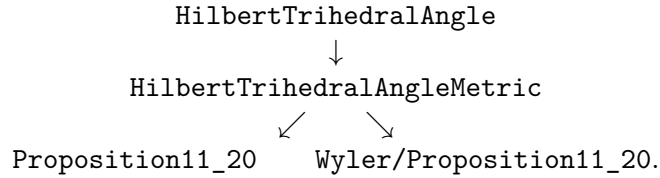
4. Expected E^4 and E^n analogue. No local 3-flat is required. The contradiction takes place inside a single ordinary plane. Therefore the same proof is meaningful in any ambient E^n satisfying the common incidence core and the same planar metric theory.

5. Missing higher-dimensional lemma. For XI.20 there is none. Neither Wyler I.7, Smith I.5, span exchange, nor a special E^4 dimension axiom is needed by the proof.

This is stronger than a reduction of XI.20 to a reconstructed local three-dimensional subspace. The proposition is, in the formal sense exposed by the proof, *dimension-free above the plane level*.

11 Dependency summary

The logical architecture of the formalization is



The shared metric module depends on the existing planar theory, in particular the formalizations of Euclid I.20 and I.25. The direct branch adds `HilbertSpaceIncidence`; the Wyler branch adds `HilbertWylerAxioms`. The final statements are intentionally separate, so the Wyler theorem is not a wrapper around the Hilbert theorem.

12 Conclusion

The formal reconstruction of XI.20 separates Euclid’s argument into an unexpectedly small spatial shell around a planar metric proof. The decisive formal device is the synthetic relation `HilbertTwoAnglesGreaterThanAngle`, which expresses Euclid’s comparison without numerical angle addition. Once the target angle is decomposed by an interior ray, the classical chain $\text{SAS} \rightarrow \text{I.20} \rightarrow \text{I.25}$ is reproduced directly.

The comparison of the two production routes yields a particularly useful structural result for the wider project: XI.20 does not depend on a specifically three-dimensional theorem. Its only ambient-space requirement is the common point–line–plane incidence core. Consequently XI.20 should transfer to E^4 and, more generally, to E^n without introducing a local 3-flat or additional dimension axioms.